Package ‘manMetaVAR’

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Title Multivariate Meta-Analysis of Vector Autoregressive Model Coefficients

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`https://jeksterslab.github.io/manMetaVAR/`,
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BugReports `https://github.com/jeksterslab/manMetaVAR/issues`

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Contents

`Compress` *Compress Replication*

Description

Compress Replication

Usage

```
Compress(taskid, repid, output_folder)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`Dynamics` *Process Dynamics*

Description

Process Dynamics

Usage

```
Dynamics(dynamics)
```

Arguments

`dynamics`	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

See Also

Other Data Generation Functions: `GenData()`

Examples

```
# stable reciprocal regulation,
Dynamics(dynamics = 1)
# escalating co-activation
Dynamics(dynamics = 2)
# adaptive recovery
Dynamics(dynamics = 3)
```

FigBias *Plot Relative Bias*

Description

Plot relative bias for the fixed and random effects.

Usage

```
FigBias(
  results,
  bias = TRUE,
  dynamics = 1,
  method = c("MetaVAR", "SeqVAR", "BMLVAR"),
  zero = FALSE,
  ylim = c(-0.15, 0.15)
)
```

Arguments

`results`	`results` data frame.
`bias`	Logical. If `bias = TRUE`, plot absolute bias. If `bias = FALSE`, plot relative bias. Note that when parameter value is equal to zero, absolute bias is used instead of relative bias.
`dynamics`	`1`, `2`, or `3`. `1` for stable reciprocal regulation, `2` for escalating co-activation, and `3` for adaptive recovery.
`method`	Character vector. Methods to include in the plot.
`zero`	Logical. Remove cases where the population parameter is equal to zero.
`ylim`	Numeric vector of length 2. Lower and upper limits for the y-axis.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: `FigCoverage()`, `FigPower()`, `FigRMSE()`, `FigType1Error()`

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigBias(results)

## End(Not run)
```

`FigCoverage` *Plot Coverage Probabilities*

Description

Plot coverage probabilities for the fixed and random effects.

Usage

```
FigCoverage(
  results,
  dynamics = 1,
  method = c("MetaVAR", "SeqVAR", "BMLVAR"),
  zero = FALSE,
  ylim = c(0, 1)
)
```

Arguments

`results`	`results` data frame.
`dynamics`	`1`, `2`, or `3`. `1` for stable reciprocal regulation, `2` for escalating co-activation, and `3` for adaptive recovery.
`method`	Character vector. Methods to include in the plot.
`zero`	Logical. Remove cases where the population parameter is equal to zero.
`ylim`	Numeric vector of length 2. Lower and upper limits for the y-axis.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: `FigBias()`, `FigPower()`, `FigRMSE()`, `FigType1Error()`

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigCoverage(results)

## End(Not run)
```

FigPower *Plot Statistical Power*

Description

Plot statistical power probabilities for the fixed and random effects.

Usage

```
FigPower(
  results,
  dynamics = 1,
  method = c("MetaVAR", "SeqVAR", "BMLVAR"),
  zero = FALSE,
  ylim = c(0, 1)
)
```

Arguments

`results`	`results` data frame.
`dynamics`	`1`, `2`, or `3`. `1` for stable reciprocal regulation, `2` for escalating co-activation, and `3` for adaptive recovery.
`method`	Character vector. Methods to include in the plot.
`zero`	Logical. Remove cases where the population parameter is equal to zero.
`ylim`	Numeric vector of length 2. Lower and upper limits for the y-axis.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: `FigBias()`, `FigCoverage()`, `FigRMSE()`, `FigType1Error()`

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigPower(results)

## End(Not run)
```

FigRMSE *Plot RMSE*

Description

Plot RMSE for the fixed and random effects.

Usage

```
FigRMSE(
  results,
  dynamics = 1,
  method = c("MetaVAR", "SeqVAR", "BMLVAR"),
  zero = FALSE,
  ylim = c(0, 0.25)
)
```

Arguments

`results`	`results` data frame.
`dynamics`	`1`, `2`, or `3`. `1` for stable reciprocal regulation, `2` for escalating co-activation, and `3` for adaptive recovery.
`method`	Character vector. Methods to include in the plot.
`zero`	Logical. Remove cases where the population parameter is equal to zero.
`ylim`	Numeric vector of length 2. Lower and upper limits for the y-axis.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: `FigBias()`, `FigCoverage()`, `FigPower()`, `FigType1Error()`

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigRMSE(results)

## End(Not run)
```

FigType1Error *Plot Type 1 Error Rates*

Description

Plot type 1 error rates for zero random effects.

Usage

```
FigType1Error(
  results,
  dynamics = 1,
  method = c("MetaVAR", "SeqVAR", "BMLVAR"),
  ylim = c(0, 0.25)
)
```

Arguments

`results`	`results` data frame.
`dynamics`	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
`method`	Character vector. Methods to include in the plot.
`ylim`	Numeric vector of length 2. Lower and upper limits for the y-axis.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: `FigBias()`, `FigCoverage()`, `FigPower()`, `FigRMSE()`

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigType1Error(results)

## End(Not run)
```

FitDTVAR — *Fit the Model using the fitDTVARMxID Package*

Description

The function fits the model using the fitDTVARMxID::fitDTVARMxID package.

Usage

```
FitDTVAR(data, wd = ".", ncores = NULL, seed = NULL)
```

Arguments

data	R object. Output of the GenData() function.
wd	Character string. Working directory.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: FitMLVAR(), FitMetaVAR(), FitMplus(), MplusInput()

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

FitMetaVAR *Multivariate Meta-Analysis using the metaVAR Package*

Description

The function performs multivariate meta-snalysis using the metaVAR::metaVAR package.

Usage

```
FitMetaVAR(data, fit, ncores = NULL)
```

Arguments

`data`	R object. Output of the `GenData()` function.
`fit`	R object. Output of the `FitDTVAR()` function.
`ncores`	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitMLVAR()`, `FitMplus()`, `MplusInput()`

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
pooled <- FitMetaVAR(
  data = data,
  fit = fit,
  ncores = parallel::detectCores()
)
summary(pooled)
print(pooled)
coef(pooled)
vcov(pooled)

## End(Not run)
```

FitMLVAR — *Fit the Model using the mlVAR Package*

Description

The function fits the model using the mlVAR::mlVAR package.

Usage

```
FitMLVAR(data, ncores = NULL)
```

Arguments

`data`	R object. Output of the `GenData()` function.
`ncores`	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitMetaVAR()`, `FitMplus()`, `MplusInput()`

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMLVAR(data = data)
print(fit)
summary(fit)

## End(Not run)
```

FitMplus — *Fit the Model using Mplus*

Description

The function fits the model using Mplus.

Usage

```
FitMplus(
  data,
  chains = 2L,
  iter = 120000L,
  fscores = NULL,
  plot = FALSE,
  default_priors = TRUE,
  wd = ".",
  mplus_bin = NULL,
  ncores = NULL
)
```

Arguments

`data`	R object. Output of the `GenData()` function.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.
`default_priors`	Logical. If `default_priors = TRUE`, use default priors.
`wd`	Character string. Working directory.
`mplus_bin`	Character string. Path to Mplus binary. If `mplus_bin = NULL`, the function will try to find the appropriate binary.
`ncores`	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitMLVAR()`, `FitMetaVAR()`, `MplusInput()`

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMplus(data = data)
print(fit)
summary(fit)

## End(Not run)
```

GenData — *Simulate Data*

Description

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

Usage

```
GenData(taskid)
```

Arguments

`taskid`	Positive integer. Task ID.

See Also

Other Data Generation Functions: `Dynamics()`

Examples

```
set.seed(42)
data <- GenData(taskid = 1)
print(data)
summary(data)
plot(data)
```

manmetavar-data-methods — *Methods for Objects of Class* `manmetavar.data`

Description

This page documents the available methods for objects of class `manmetavar.data`.

Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)

## S3 method for class 'manmetavar.data'
summary(object, ...)

## S3 method for class 'manmetavar.data'
plot(x, ...)
```

Arguments

`x`	Object of class `manmetavar.data`.
`...`	additional arguments.
`object`	Object of class `manmetavar.data`.

Author(s)

Ivan Jacob Agaloos Pesigan

`manmetavar-dtvar-methods`

Methods for Objects of Class `manmetavar.dtvar`

Description

This page documents the available methods for objects of class `manmetavar.dtvar`.

Usage

```
## S3 method for class 'manmetavar.dtvar'
coef(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  ...
)

## S3 method for class 'manmetavar.dtvar'
vcov(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
```

```
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  ...
)

## S3 method for class 'manmetavar.dtvar'
print(
  x,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.dtvar'
summary(
  object,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)
```

Arguments

`object`	Object of class `manmetavar.dtvar`.
`alpha`	Logical. If `alpha = TRUE`, include estimates of the `alpha` vector, if available. If `alpha = FALSE`, exclude estimates of the `alpha` vector.
`beta`	Logical. If `beta = TRUE`, include estimates of the `beta` matrix, if available. If `beta = FALSE`, exclude estimates of the `beta` matrix.

nu	Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.
psi	Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.
theta	Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix.
converged	Logical. Only include converged cases.
grad_tol	Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.
hess_tol	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE.
vanishing_theta	Logical. Test for measurement error variance going to zero if converged = TRUE.
theta_tol	Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.
...	additional arguments.
x	Object of class manmetavar.dtvar.
means	Logical. If means = TRUE, return means. Otherwise, the function returns raw estimates.
digits	Integer indicating the number of decimal places to display.

Author(s)

Ivan Jacob Agaloos Pesigan

---

manmetavar-metavar-methods

Methods for Objects of Class manmetavar.metavar

Description

This page documents the available methods for objects of class manmetavar.metavar.

Usage

```
## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.metavar'
vcov(object, ...)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = 0.05, digits = 4, ...)
```

```
## S3 method for class 'manmetavar.metavar'
summary(object, alpha = 0.05, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, lb = TRUE, ...)
```

Arguments

`object`	Object of class `manmetavar.metavar`.
`...`	additional arguments.
`x`	Object of class `manmetavar.metavar`.
`alpha`	Numeric vector. Significance level α.
`digits`	Integer indicating the number of decimal places to display.
`parm`	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
`level`	the confidence level required.
`lb`	Logical. If `TRUE`, returns profile likelihood-based confidence intervals. If `FALSE`, returns Wald confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

`manmetavar-mlvar-methods`

Methods for Objects of Class `manmetavar.mlvar`

Description

This page documents the available methods for objects of class `manmetavar.mlvar`.

Usage

```
## S3 method for class 'manmetavar.mlvar'
print(
  x,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.mlvar'
summary(
```

```
  object,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)
```

Arguments

`x`	Object of class `manmetavar.mlvar`.
`show`	Tables to show.
`digits`	Integer indicating the number of decimal places to display.
`...`	additional arguments.
`object`	Object of class `manmetavar.mlvar`.

Author(s)

Ivan Jacob Agaloos Pesigan

`manmetavar-mplus-methods`

Methods for Objects of Class `manmetavar.mplus`

Description

This page documents the available methods for objects of class `manmetavar.mplus`.

Usage

```
## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
plot(
  x,
  what = "posterior",
  parm = NULL,
  level = 0.95,
```

```
  burnin = NULL,
  legend_loc = "topright",
  ...
)
```

Arguments

`object`	Object of class `manmetavar.mplus`.
`median`	Logical. If `median = TRUE`, return median of the posterior. If `median = FALSE`, return mean of the posterior.
`burnin`	Integer indicating initial samples to discard. If `burnin = NULL`, do not discard anything.
`...`	additional arguments.
`alpha`	Numeric vector. Significance level α.
`digits`	Integer indicating the number of decimal places to display.
`parm`	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
`level`	the confidence level required.
`x`	Object of class `manmetavar.mplus`.
`what`	Character string. If `what = "posterior"`, return posterior distribution plots. If `what = "trace"`, return trace plots.
`legend_loc`	Character string. Legend location.

Author(s)

Ivan Jacob Agaloos Pesigan

`MplusInput` *Create Mplus Input File*

Description

The function creates an Mplus input file.

Usage

```
MplusInput(
  dynamics,
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,
```

```
  fscores,
  plot,
  default_priors,
  ncores = NULL
)
```

Arguments

dynamics	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
fn_data	Character string. Filename for data file.
fn_posterior	Character string. Filename for posterior output.
fn_factorscores	Character string. Filename for factor scores output.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.
default_priors	Logical. If `default_priors = TRUE`, use default priors.
ncores	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitMLVAR()`, `FitMetaVAR()`, `FitMplus()`

Examples

```
cat(
  MplusInput(
    dynamics = 1,
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
    chains = 2L,
    iter = 60000L,
    fscores = 1000L,
    plot = TRUE,
    default_priors = TRUE,
    ncores = NULL
  )
)
```

`params` *Simulation Parameters*

Description

Simulation Parameters

Usage

```
data(params)
```

Format

A dataframe with 27 rows and 4 columns:

taskid Simulation Task ID.

n Sample size.

time Number of measurement occassions.

dynamics Process dynamics. `1` for stable reciprocal regulation, `2` for escalating co-activation, and `3` for adaptive recovery.

Author(s)

Ivan Jacob Agaloos Pesigan

`results` *Simulation Results*

Description

Simulation Results

Usage

```
data(results)
```

Format

A dataframe with I rows and J columns:

taskid Simulation Task ID.

replications Number of replications.

parnames Parameter names.

parameter Population parameter value.

method Method used to fit the data.

n Sample size.

time Number of measurement occassions.

dynamics Process dynamics. `1` for stable reciprocal regulation, `2` for escalating co-activation, and `3` for adaptive recovery.

ci Confidence/credible intervals method.

est Mean parameter estimate.

se Mean standard error.

t Mean test statistic.

p Mean p-value.

ll Mean lower limit of the 95% confidence/credible interval.

ul Mean upper limit of the 95% confidence/credible interval.

sig Proportion of statistically significant results.

theta_hit Proportion of replications where the confidence intervals contained the population parameter.

sq_error Mean squared error.

bias Bias.

rel_bias Relative bias.

se_bias Bias in standard error estimate.

rel_se_bias Relative bias in standard error estimate.

rmse Root mean square error.

coverage Coverage probability.

power Statistical power.

par_idx Parameter index.

Author(s)

Ivan Jacob Agaloos Pesigan

`Sim` *Simulation Replication*

Description

Simulation Replication

Usage

```
Sim(
  taskid,
  repid,
  output_folder,
  overwrite,
  integrity,
  seed,
  data,
  metavar,
  mlvar,
  mplus,
  chains,
  iter,
  fscores,
  plot,
  default_priors
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`seed`	Integer. Random seed.
`data`	Logical. Simulate data.
`metavar`	Logical. If `metavar = TRUE`, fit the model using the `metaVAR` package.
`mlvar`	Logical. If `mlvar = TRUE`, fit the model using the `mlVAR` package.
`mplus`	Logical. If `mplus = TRUE`, fit the model using DSEM in `Mplus`.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.
`default_priors`	Logical. If `default_priors = TRUE`, use default priors.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitDTVAR` *Simulation Replication - FitDTVAR*

Description

Simulation Replication - FitDTVAR

Usage

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitMetaVAR` *Simulation Replication - FitMetaVAR*

Description

Simulation Replication - FitMetaVAR

Usage

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMLVAR — *Simulation Replication - FitMLVAR*

Description

Simulation Replication - FitMLVAR

Usage

```
SimFitMLVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplus *Simulation Replication - FitMplus*

Description

Simulation Replication - FitMplus

Usage

```
SimFitMplus(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot,
  default_priors
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.
`default_priors`	Logical. If `default_priors = TRUE`, use default priors.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFN *Simulation File Name*

Description

Simulation File Name

Usage

```
SimFN(output_type, output_folder, suffix)
```

Arguments

`output_type`	Character string. Output type. Valid values include `"data"`, `"fit-dt-var-mx"`, `"fit-meta-var-mx"`, and `"fit-ml-var"`.
`output_folder`	Character string. Output folder.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.

Value

Returns a character string file name with the `output_folder` in the OS-specific format.

SimGenData *Simulation Replication - GenData*

Description

Simulation Replication - GenData

Usage

```
SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimProj — *Simulation Project Name*

Description

Simulation Project Name

Usage

```
SimProj()
```

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

`Sum` *Summary*

Description

Summary

Usage

```
Sum(taskid, reps, output_folder, overwrite, integrity, metavar, mlvar, mplus)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`metavar`	Logical. If `metavar = TRUE`, fit the model using the `metaVAR` package.
`mlvar`	Logical. If `mlvar = TRUE`, fit the model using the `mlVAR` package.
`mplus`	Logical. If `mplus = TRUE`, fit the model using DSEM in `Mplus`.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumFitMetaVARLB` *Summary: Likelihood-Based (FitMetaVAR)*

Description

Summary: Likelihood-Based (FitMetaVAR)

Usage

```
SumFitMetaVARLB(taskid, reps, output_folder, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumFitMetaVARNormal` *Summary: Normal Theory (FitMetaVAR)*

Description

Summary: Normal Theory (FitMetaVAR)

Usage

```
SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMLVAR *Summary (FitMLVAR)*

Description

Summary (FitMLVAR)

Usage

```
SumFitMLVAR(taskid, reps, output_folder, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMplus *Summary (FitMplus)*

Description

Summary (FitMplus)

Usage

```
SumFitMplus(taskid, reps, output_folder, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

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